

Lab 1



Prefix Averages

- Implement the prefix average algorithm with linear time complexity

Algorithm *prefixAverages2*(X, n)

Input array X of n integers

Output array A of prefix averages of X

$A \leftarrow$ new array of n integers

$s \leftarrow 0$

for $i \leftarrow 0$ to $n - 1$ **do**

$s \leftarrow s + X[i]$

$A[i] \leftarrow s / (i + 1)$

return A

Maximum Subsequence Sum Problem

Given a set of integers $A_1, A_2, \dots, A_N$, find the maximum value of

$$\sum_{k=i}^j A_k$$

For convenience, the maximum subsequence sum is zero if all the integers are negative.

```
int MaxSubSum2(int *a, int total) {  
    int maxSum=0;  
  
    for (int i=0; i<total; i++) {  
        thisSum=0;  
        for (int j=i; j<total; j++) {  
            thisSum+=a[j];  
  
            if (thisSum>maxSum)  
                maxSum=thisSum;  
        }  
    }  
    return maxSum;  
}
```

$O(n^2)$

Lab 1 Submission

Due at 23:59, March 25.

Note: Submission after 23:59, March 25 will NOT be graded.

Zip file name: StudentID_LastName_FirstName_Lab#
(Please use Zip to compress your submission)

Replace # with the lab number.

Assume my ID is 12345678, the zip file name for Lab 1 is:

12345678_Liang_Pengpeng_Lab1

Note: If you do not follow the rule of the name of the Zip file, I might not be able to find your submission.

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